Homework #1; 0701432

Q1) $f(x) = x^{-2}$. $a = 1$

(a) $P_4(x) = ?$

$$P_4(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2}(x-1)^2 + \frac{f^{(3)}(1)}{6}(x-1)^3 + \frac{f^{(4)}(1)}{24}(x-1)^4$$

$f'(x) = -2x^{-3}$; $f''(x) = 6x^{-4}$; $f^{(3)}(x) = -24x^{-5}$; $f^{(4)}(x) = 120x^{-6}$; $f^{(5)}(x) = -720x^{-7}$

$f(1) = 1$; $f'(1) = -2$; $f''(1) = 6$; $f^{(3)}(1) = -24$; $f^{(4)}(1) = 120$

$$\Rightarrow P_4(x) = 1 - 2(x-1) + 3(x-1)^2 - 4(x-1)^3 + 5(x-1)^4$$

(b) $f(0.9) \approx P_4(0.9) = 1 + 0.2 + 0.03 + 0.004 + 0.0005$
$= 1.2345$

$f(1.1) \approx P_4(1.1) = 1 - 0.2 + 0.03 - 0.004 + 0.0005$
$= 0.8265$

(c) Error $= R_4(x) = \frac{f^{(5)}(c)}{120}(x-1)^5 = \frac{-6}{c^7}(x-1)^5$, c between 1 and x

Error bound:

for $f(0.9) \approx P_4(0.9)$

$$|\text{Error}| = \frac{6}{c^7}(0.1)^5 < \frac{6}{(0.9)^7}(0.1)^5 = 0.0001254$$
$$= 1.254 \times 10^{-4}$$

for $f(1.1) \approx P_4(1.1)$

$$|\text{Error}| = \frac{6}{c^7}(0.1)^5 < 6(0.1)^5 = 0.00006$$
$$= 0.6 \times 10^{-4}$$

The approximation $f(1.1) \approx P_4(1.1)$ is closer to the correct value

(d) for $x = 0.9$, The actual error $= |P_4(0.9) - f(0.9)|$

$$= |1.2345 - 1.2345679|$$
$$= 0.0000679 = 0.679 \times 10^{-4}$$

for $x = 1.1$, The actual error $= |P_4(1.1) - f(1.1)| = |0.8265 - 0.8264463|$
$= 0.0000537 = 0.537 \times 10^{-4}$

Q2) (a) $P(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \frac{f^{(3)}(0)}{6}x^3 + \frac{f^{(4)}(0)}{24}x^4 + \frac{f^{(5)}(0)}{120}x^5$

$f(x) = \cos x;\ f'(x) = -\sin x;\ f''(x) = -\cos x;\ f^{(3)}(x) = \sin x;\ f^{(4)}(x) = \cos x;\ f^{(5)}(x) = -\sin x$

$f(0) = 1,\ f'(0) = 0,\ f''(0) = -1,\ f^{(3)}(0) = 0,\ f^{(4)}(0) = 1,\ f^{(5)}(0) = 0$

$\Rightarrow P(x) = P_5(x) = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$

(b) Error $= R_5(x) = \frac{f^{(6)}(c)}{720}x^6$, c lies between 0 and x

$$|\text{Error}| = \frac{|-\cos c|}{720}|x|^6 = \frac{|\cos c|}{720}|x|^6$$

$$\leq \frac{1}{720}|x|^6$$

$x \in [-\pi/4, \pi/4] \Rightarrow |x| \leq \pi/4$

$$\Rightarrow |\text{Error}| \leq \frac{1}{720}\left(\frac{\pi}{4}\right)^6 = 0.000326$$

$$= 0.326 \times 10^{-3}$$

Q3) (a) $s = 0$ ($\equiv$ Positive)

$c = 1000001010 = 2^1 + 2^3 + 2^{10} = 1034$

$p = c - \text{bias} = c - 1023 = 1034 - 1023 = 11$

$f = 0.1001001100\cdots0 = \frac{1}{2} + \frac{1}{2^4} + \frac{1}{2^7} + \frac{1}{2^8}$

$= \frac{1}{2} + \frac{1}{16} + \frac{1}{128} + \frac{1}{256}$

$= \frac{128 + 16 + 2 + 1}{256}$

$= \frac{147}{256}$

The normalized floating-point form is

$$+1.\underbrace{\boxed{1001001100\cdots0}}_{52\text{ bits}} \times 2^{11}$$

Its decimal equivalent is

$$+\left(1 + \frac{1}{2} + \frac{1}{2^4} + \frac{1}{2^7} + \frac{1}{2^8}\right) \times 2^{11}$$

$$= +\left(2^{11} + 2^{10} + 2^7 + 2^4 + 2^3\right)$$

$$= +\left(2048 + 1024 + 128 + 16 + 8\right)$$

$$= +3224$$

(b) $s=1$ (= negative)

$c = 10000000010 = 2+2^3+2^{10} = 1034$

$p = c - \text{bias} = c - 1023 = 1034 - 1023 = 11$

$f = 0.100100110 0 \cdots 0 = \frac{1}{2} + \frac{1}{2^4} + \frac{1}{2^7} + \frac{1}{2^8}$

$\Rightarrow$ The normalized floating-point form is

$$-1.\underbrace{\boxed{100100110 0 \cdots 0}}_{52\text{ bits}} \times 2^{11}$$

decimal

its equivalent is

$$-\left(1 + \frac{1}{2} + \frac{1}{2^4} + \frac{1}{2^7} + \frac{1}{2^8}\right) \times 2^{11}$$

$$= -3224$$

(c) $s=0$ (= positive)

$c = 0111111111 1 = 1+2+2^2+2^3+2^4+2^5+2^6+2^7+2^8+2^9$

$$= \frac{1-2^{10}}{1-2} = 2^{10}-1 = 1023$$

$p = c - \text{bias} = c - 1023 = 1023 - 1023 = 0$

$f = 0.0101001100 \cdots 0 = \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^7} + \frac{1}{2^8}$

$\Rightarrow$ The normalized floating-point form is

$$+1.\underbrace{\boxed{0101 0011 00 \cdots 0}}_{52\text{ bits}} \times 2^0$$

its decimal equivalent is

$$+\left(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{128} + \frac{1}{256}\right) = +\frac{(256+64+16+2+1)}{256}$$

$$= +\frac{339}{256}$$

$$= +1.32421875$$

(d) $s = 0$ (= positive)

$c = 01111111111 = 1023$

$p = c - \text{bias} = c - 1023 = 1023 - 1023 = 0.$

$f = 0.0101001100\cdots01$

$= \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^7} + \frac{1}{2^8} + \cdots \frac{1}{2^{52}}$

$\Rightarrow$ The normalized floating-point form is

$$+1.\underbrace{\boxed{0101001100\cdots01}}_{52\text{ bits}} \times 2^0$$

Its decimal equivalent is

$$+\left(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{128} + \frac{1}{256} + \frac{1}{2^{52}}\right)$$

$$= +\left(\frac{339}{256} + \frac{1}{2^{52}}\right) = \frac{(339) \times 2^{44} + 1}{2^{52}}$$

$$= 1.32421875000000022204460492503130808\cdots$$

Q4) (a) The next largest

$$0\,10000001010\;10010011\underbrace{000\cdots0}_{43\text{ zero}}1$$

The next smaller

$$0\,10000001010\;10010010\underbrace{111\cdots11}_{44\text{ one}}$$

(b), (c), (d) Similar to (a)

Q5)

$$\left.\begin{aligned} \tfrac{2}{3}\times 2 &= \tfrac{1}{3}+1 \\ \tfrac{1}{3}\times 2 &= \tfrac{2}{3}+0 \end{aligned}\right\}\text{ repeated}$$

$$\tfrac{2}{3}\times 2$$

$$\Rightarrow \tfrac{2}{3} = 0.\overline{10}$$

Thus $\frac{2}{3} = +1.0\overline{10} \times 2^{-1}$

$$= +1.0\,1010101010\cdots \times 2^{-1}$$

$$= +1.\boxed{0\,\underbrace{1010\cdots10}_{25\text{ collection}}}\,1010\cdots \times 2^{-1}$$

52 bits

$$\Rightarrow fl\left(\tfrac{2}{3}\right) = +1.0\,\underbrace{1010\cdots101}_{25\text{ collection}} \times 2^{-1}$$

Now $fl\left(\frac{2}{3}\right) = \frac{2}{3} - \text{Tail}$

$$\text{Tail} = 0.01010101010\cdots \times 2^{-52} \times 2^{-1}$$

$$= 0.\overline{01} \times 2^{-52} \times 2^{-1}$$

$$x = 0.\overline{01} \Rightarrow 2^2 x = 01.\overline{01}$$

$$2^2 x - x = 01 = 1 \Rightarrow 3x = 1 \Rightarrow x = 1/3$$

$$\Rightarrow \left|fl(2/3) - \tfrac{2}{3}\right| = \tfrac{1}{3}\times 2^{-1} \times 2^{-52} = \tfrac{1}{6}\times 2^{-52}$$

Q5) $9.6 = (?)_2$

$9 = 1001.$

$$\left.\begin{array}{l} 0.6 \times 2 = 0.2 + 1 \\ 0.2 \times 2 = 0.4 + 0 \\ 0.4 \times 2 = 0.8 + 0 \\ 0.8 \times 2 = 0.6 + 1 \end{array}\right\} \text{repeated}$$

0.6×2

$\Rightarrow 0.6 = 0.\overline{1001}$

$$\Rightarrow 9.6 = 1001.\overline{1001} = 1.001\,\overline{1001} \times 2^3$$

$$= +1.001\,1001\,1001\,1001\cdots \times 2^3$$

$$= +1.001\,\boxed{\underbrace{1001\,1001\cdots1001}_{12 \text{ collection}}}\,1001\cdots \times 2^3$$

52 bits

$$\Rightarrow fl(9.6) = +1.001\,\underbrace{1001\,1001\cdots1001}_{12 \text{ collection}}\,1 \times 2^3$$

Now $fl(9.6) = 9.6 - \text{Tail}$

$$\text{Tail} = 0.001\,1001\,1001\cdots \times 2^{-52} \times 2^3$$

$$= 0.\overline{0011} \times 2^{-52} \times 2^3$$

$$x = 0.\overline{0011} \Rightarrow 2^4 x = 0011.\overline{0011}$$

$$2^4 x - x = 0011 = 3$$

$$15x = 3 \Rightarrow x = \tfrac{1}{5}$$

$$\Rightarrow |fl(9.6) - 9.6| = \frac{8}{5} \times 2^{-52}$$

Q6) (a) $1.\boxed{00\cdots\cdots 0} \times 2^0$

$+\ 0.\boxed{00\cdots\cdots 0011}01 \times 2^0$

$1.\boxed{00\cdots\cdots 0011}01 \times 2^0$

saved as

$1.00\cdots\cdots 00\cancel{01}\cancel{1} \times 2^0$

when we subtract 1 we get

$0.00\cdots\cdots 00\cancel{01}\cancel{1} \times 2^0$

$= 2^{-51} + 2^{-52} = 2\,2^{-52} + 2^{-52} = 2^{-52}(3) = 3\,\epsilon_{mach}$

(b) $1.\boxed{00\cdots\cdots 0} \times 2^0$

$+\ 0.\boxed{00\cdots\cdots 011}\cdots 1^{\text{bit (60)}} \times 2^0$

$1.\boxed{00\cdots\cdots 011}00\cdots 01 \times 2^0$

bit 60

Saved as $1.\boxed{00\cdots\cdots 011} \times 2^0$

when we subtract 1 we get

$0.00\cdots\cdots 011 \times 2^0$

$= 2^{-51} + 2^{-52}$

$= 3\,(2^{-52}) = 3\,\epsilon_{mach}$